# Assemblies, constraints & file types

GRAPHICS WITH MR KELLY

Higher · CAD & technology · Worksheet

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ assemble components with mate, align and centre axis
- ☐ use orientate, offset and tangent constraints to fully constrain an assembly
- ☐ explain top-down and bottom-up modelling and sub-assemblies
- ☐ explain STEP, IGES, DXF and 3DS file types

## I DO — Worked example

Read how a full-mark answer is built

*A wheel is fitted onto an axle and pushed against the side of a bracket, with its outer face level with the bracket's outer face.*

**1 State the three assembly constraints used.** (3 marks)

**Full-mark answer** each ✓ is one mark

Centre axis, to line the wheel hole up with the axle ✓. Mate, to bring the wheel face against the bracket ✓. Align, to make the outer faces level ✓.

**Why marks are not awarded**

*"Line up, push together and make level."*

0 marks: the course specification terms must be used.

## WE DO — Guided practice

Use the starters and word bank to help you

*An engineer produces orthographic drawings directly from a 3D CAD assembly.*

**2 Describe two advantages of producing the drawings from the 3D model.** (2 marks)

**Sentence starters**

One … is …, which means …

Another … is …, so …

**Word bank**

always · straight · accurate · update · changed · automatically · drawings · dimensions

Tip: 2 marks = 2 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** Which constraint lines up a pin with a hole?
☐ Mate ☐ Align ☐ Centre axis ☐ Fillet

**b)** Two faces level with each other, facing the same way, is...
☐ Align ☐ Mate ☐ Centre axis ☐ Chamfer

**c)** Which format would you send to a company using different CAD software?
☐ JPG ☐ STEP ☐ DOCX ☐ PNG

**d)** Modelling parts inside the assembly so changes update related parts is...
☐ Bottom-up ☐ Exploded ☐ Top-down ☐ Suppressed

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Mate** | ____ | **A** Two faces brought together facing each other. |
| 2. **Align** | ____ | **B** Two faces made flush, facing the same way. |
| 3. **Centre axis** | ____ | **C** Lines up the axes of two round features. |
| 4. **Orientate** | ____ | **D** Parts modelled in the assembly referencing each other; changes update automatically. |
| 5. **Offset constraint** | ____ | **E** Sets an angle between two parts. |
| 6. **Top-down modelling** | ____ | **F** Sets a distance between two faces. |

*A needle cartridge is modelled in position by projecting the hole sizes from the tonearm head (top-down modelling).*

### 5 Explain two benefits of modelling in this way. (2 marks)

*A supplier sends a 3D model of a motor to a company that uses different CAD software.*

### 6 State a suitable file format and explain why it is used. (2 marks)

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*A tonearm assembly has a base, an axle and a main arm. The arm pivots on the axle and is set at 20° to the base, 13 from the base's inner edge.*

**7 Describe how the tonearm would be fully assembled.** (4 marks)

**Check and reflect**

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?